



Enterprise Incident Response & Technical Forensic Report

Multi-Stage Privilege Escalation & Lateral Movement Breach Management

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FRAMEWORK FOCUS:	NIST SP 800-61 Rev. 2 Framework & PCI-DSS v4.0 Compliance
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1. Executive Summary & Core Performance Metrics

On Tuesday at 03:14 AM WAT, a sophisticated threat actor executed an unauthorized intrusion into the NairaFlow Technologies corporate infrastructure. The vector initiated via a successful phishing vector targeting a developer’s local workstation, leading to the extraction of long-lived AWS IAM configuration keys. The attacker utilized "living-off-the-land" methodologies to move horizontally from the Staging VPC across unsegmented boundaries directly into the Production VPC infrastructure, attempting to query production customer financial ledgers and masked Personally Identifiable Information (PII).

In accordance with the **NIST SP 800-61 Rev. 2** guidelines, the Incident Response Team immediately initiated containment scripts, safeguarding core processing workflows and preventing systematic exfiltration.

 **REAL-TIME SECURITY RESPONSE KPIS**

Operational Metric	Compliance Target	Incident Performance	Status
Mean Time to Containment (MTTC)	< 4.0 Hours	3 Hours, 42 Minutes	TARGET MET
Core Gateway Production Uptime	≥ 99.9%	99.94%	TARGET MET
Volatile Memory & Integrity Preservation	100% Intact	100% (No Hard Reboots)	TARGET MET
Team Communication Out-of-Band	Mandatory	100% Signal App Deployment	TARGET MET

2. Evidence Preservation & Forensic Acquisition

To comply with forensic readiness criteria, traditional anti-patterns like broad instance shutdowns or server hard reboots were strictly prohibited. Volatile runtime states, container lifecycles, and cryptographic environments were kept operational to protect crucial evidentiary trails.

2.1 Forensic Chain of Custody Registry

Evidence ID	Source Host System	Evidence Description	Cryptographic SHA-256 Signature	Custodian Account
EVID-001	EKS Production Node Group	Live EBS Snapshot of worker block layer	e3b0c44298fc1c149afb4c8996fb92427ae41e...	J. C. Lawrence
EVID-002	AWS CloudTrail Engine	JSON Audit trail capture file containing API logs	4a8a08f09d37b73795649038408b5f33a94a115...	J. C. Lawrence

2.2 AWS CloudTrail Indicator of Compromise (IoC)

The following raw structural audit log details the exact vector where the malicious actor initiated an unauthorized session cross-boundary request to acquire administrative level operational authority:

```
{
  "eventVersion": "1.08",
  "userIdentity": {
    "type": "IAMUser",
    "principalId": "AIDAJQAB2M4FXXXXXXXXX",
    "arn": "arn:aws:iam::123456789012:user/nairaflow-dev-staging",
    "accountId": "123456789012",
    "accessKeyId": "AKIAIOSFODNN7EXAMPLE",
    "userName": "nairaflow-dev-staging"
  },
  "eventTime": "2026-06-16T03:14:22Z",
  "eventSource": "sts.amazonaws.com",
  "eventName": "AssumeRole",
  "awsRegion": "eu-west-1",
  "sourceIPAddress": "10.0.45.12",
  "requestParameters": {
    "roleArn": "arn:aws:iam::123456789012:role/NairaFlow-Prod-RDS-FullAccess",
    "roleSessionName": "exfiltrator-session"
  }
}
```

3. Containment & Eradication Engineering

Surgical orchestration isolated the threat vector without breaking the core transaction gateway's capability to process legitimate flows.

3.1 Out-of-Band Incident Communication Strategy

As a foundational defensive action against potential enterprise directory or Slack visibility compromise by the adversary, all operational interactions shifted to an end-to-end encrypted out-of-band communication mesh via **Signal**.

3.2 Immediate Credential Invalidation Script

The following script applied a hard policy block across the targeted credentials without shutting down active computing runtimes:

```
#!/bin/bash
# Surgical containment of compromised staging keys without impacting production clusters
TARGET_USER="nairaflow-dev-staging"
POLICY_NAME="ImmediateContainmentDeny"

aws iam put-user-policy --user-name $TARGET_USER --policy-name $POLICY_NAME --policy-document '{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Deny",
      "Action": "*",
      "Resource": "*"
    }
  ]
}'
```

4. Post-Incident Remediation & Hardening Roadmap

To align with standard compliance frameworks like **PCI-DSS v4.0** and **ISO 27001**, a structural modernization plan was drafted to eliminate vulnerabilities.

4.1 Kubernetes Micro-segmentation Rule

This manifest eliminates cross-namespace horizontal movement vectors permanently by denying cross-boundary connectivity between Staging and Production:

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: deny-staging-to-production
  namespace: production
spec:
  podSelector: {}
  policyTypes:
  - Ingress
  ingress:
  - from:
    - namespaceSelector:
        matchLabels:
          kubernetes.io/metadata.name: production
```

4.2 Long-Term Strategic Hardening Actions

- **Automated Secret Lifecycles:** Integration of HashiCorp Vault to automate secret rotations.
- **Attack Surface Optimization:** Removal of long-lived access parameters to shrink the IAM permission area surface by $\Delta = 75\%$.